

Application of Bayesian Network in Information Fusion Analysis of Four Diagnostic Methods of Traditional Chinese Medicine

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Abstract—Bayesian network is the effective tool for reasoning and modeling of complex and uncertain system based on traditional probability theory, which is widely used in uncertain decision-making, data analysis, intelligence reasoning and other fields. Syndrome differentiation and treatment, one of the basic characteristics of traditional Chinese medicine (TCM), is the essence of TCM. Fusion analysis on standardization and objectification of four diagnostic methods of TCM is the basic of syndrome differentiation analysis of TCM. The traditional methods are often with subjective differentiation and ambiguity, and the essence of syndrome differentiation of TCM can be seen as a classification problem. Bayesian network, as a better algorithm of data mining, is being increasingly applied to the study of syndrome differentiation of TCM. This article outlines the application of Bayesian network in information fusion analysis of four diagnostic methods of TCM and prospects for future research.

Keywords—information fusion; traditional Chinese medical diagnosis; Bayesian network

I. INTRODUCTION

Medical science is one subject of natural science and each leap of the medical science always benefits from the science and technology. In the past century, the ultrasound imaging technology, computer technology, lasers and other frontier science have pushed the Western medicine rapidly develop. But TCM has developed under a different mode due to different historical and cultural background. At present, if the TCM expects to step to the world, it must apply the modern science and technology to scientifically explain the TCM theory and practice. Syndrome differentiation and treatment, one of the basic characteristics of TCM, is the essence of TCM. The accuracy of syndrome differentiation and treatment is an important prerequisite to obtain the curative effect. However, so many diagnostic and therapeutic experiences are difficult to be repeated due to the lack of objective determination of syndromes and quantitative research methods. Therefore, the analysis on standardization of syndrome differentiation of TCM is the key to develop TCM theory, clinical efficacy and the internationalization. Fusion analysis on standardization and objectification of four diagnostic methods of TCM is the basic of syndrome

differentiation analysis of TCM. The traditional methods are often with subjective differentiation and ambiguity, and the essence of syndrome differentiation of TCM can be seen as a classification problem, which introducing the data mining technology to the research of TCM and building vast databases, and also using methods of multivariate statistical analysis to make the differentiation results more objective. Bayesian network, as a better algorithm of data mining, is being increasingly applied to the study of syndrome differentiation of TCM.

II. OVERVIEW OF BAYESIAN NETWORK

A. Principle and the Structure of Bayesian Network

Bayesian network is an object directed acyclic graph model with probability comments used for uncertainty reasoning. [1] It can predict the unknown events by probabilistic methods based on prior knowledge and existing statistical data. It describes the relationships between the data through an intuitive graphical method and expresses many dependencies among multiple variables with the weight of probability measure, such as the timing relationship, correlation, causation and so on. [2] Due to its solid theoretical foundation, intuitive knowledge representation, flexible reasoning ability and facilitate decision-making mechanism, Bayesian network becomes the emerging technology in the data mining field. [3] Bayesian network is composed of representative variable nodes and the arrows connecting the nodes. Each node represents a random event variable, and the arrows represent the dependencies between nodes. Whenever the emergence of a reason node leads to a result, it should be expressed by conditional probability, not the inevitable result. The intensity of probability of the relationships between event variables constitutes the conditional probability table of Bayesian network. You can quickly get the combination probability of each basic event according to the Bayesian network structure and conditional probability table.

B. Learning Purpose and the Construction of Bayesian Network

One of the purposes of learning Bayesian network is to achieve the reasoning of Bayesian network. Under the

conditions of knowing the occurrence of some variables, computer can calculate causation between the variables and the local probability distribution of each variable from the probability tables. That is to inquire posterior probability based on prior probability. About Bayesian network structure and the prior probability distribution, the prior probability can be specified by means of human experience, expert knowledge, or the true probability distribution can be determined according to the direct analysis of the features of the data and data statistics. To establish a database based on prior knowledge and available statistical data, so that the computer can automatically learn according to Bayesian network learning algorithm, and then get Bayesian network model in corresponding areas. [4]

III. FEASIBILITY OF THE APPLICATION OF BAYESIAN NETWORK IN INFORMATION FUSION ANALYSIS OF FOUR DIAGNOSTIC METHODS OF TCM

A. The Laws of Syndrome Differentiation of TCM

Syndrome differentiation and treatment is the characteristic of TCM, and also the most obvious characteristic to distinguish the TCM from modern medicine. "Syndrome", as a unique concept in TCM, is the pathological summary of disease location, etiology, disease character and patient's condition at current stage in the disease process; the composition of pathogenic factors and body reaction; and the conclusions of current nature of the disease. Emphasizing on combined use of the four diagnostic methods, TCM gathers information through inspection, auscultation, inquiry, and palpation. And then it will do comprehensive analysis; discern disease location, etiology, disease character; and seize the fundamental mechanism of lesion through the phenomenon to search the root cause of a disease.

Patients' symptoms, signs and other clinical manifestations are known as symptoms in TCM. The purpose of syndrome differentiation in TCM is to clear and define disease location, disease character and other elements which are called the key syndrome elements for short. The mutual combination of the key syndrome elements can be summarized into a complete diagnosis of name of syndrome. Therefore, "the key syndrome element" defines the disease location and disease character through the identification of "syndrome", and constitutes the basic element of the "name of syndrome". Based on TCM theory, "syndrome differentiation" defines the pathological nature-the key syndrome elements through the analysis of syndrome, and then work out the process of thing and cognition of the diagnosis of the name of syndrome. [5]

There are about a thousand of common clinical symptoms, such as chills, fever, dizziness, headache, chest tightness, coughing, sweating, thirst, dry stool, pale complexion, purple tongue, greasy tongue coating, pulse string, etc. There are about 50 common syndrome elements, such as disease locations of heart, lung, spleen, liver, kidney, stomach, exterior, interior, etc, and diseases character of Qi deficiency, Qi stagnation, Blood deficiency, Blood stasis, deficiency of Yin, deficiency of Yang, cold, heat, dampness,

dryness, sputum, water, etc. And there are thousands of name of syndrome formed by the combination of 50 key syndrome elements, such as the Heart Yang deficiency, Liver depression and Qi stagnation, Yang deficiency of spleen and kidney, heart-kidney imbalance and so on.

The diversity of combination and evolution of key syndrome elements, form the complexity of disease, and also reflect the flexibility of syndrome differentiation in TCM. Though there are differences between physicians, their basic principle of syndrome differentiation thinking are interlinked. There are certain rules for the definition of key syndrome elements; the degree of subordination of a group of symptoms to some key syndrome elements; how the syndrome elements combined; and the category of syndrome type. [6]

B. Syndrome Differentiation of TCM and Bayesian Network

The bottleneck of the research about the standardization of syndrome differentiation in TCM is due to backward technology in collecting, storing and analyzing the vast amounts of information of the four diagnostic data. Syndrome differentiation of TCM is very complex; but with mathematical methods and computer technology such as data mining and information processing, we can find out the rule by doing quantitative analysis and identification research about the syndrome differentiation of the key syndrome element. The feature of Bayesian network model decides that it can be used to describe complex causal relationship. In fusion analysis of four diagnostic method information, Bayesian network model can be used to research the causal relationship between symptoms, symptoms and the key syndrome elements, and disease and syndrome. Structure learning and principles of reasoning of Bayesian network are also quite similar to the thinking process of the syndrome differentiation of TCM; so the method that how Bayesian network deal with uncertainty can also be used to describe the process of the syndrome differentiation in TCM, thus to do the analysis and reasoning judgment.

IV. CURRENT STATUS OF BAYESIAN NETWORK IN TCM RESEARCH

By direct analysis of relevant factors of syndrome type diagnosis of electronic cases in of Harbin Hospital of TCM HIS system, Liu Zhanqing [7] constructed Naive Bayesian and Expand Bayesian model. The result shows: Bayesian classification algorithm has good classification performance in clinical diagnosis of TCM.

Zhang Liwei [8] collected 187 chest obstruction clinical cases. 170 of the cases are randomly selected as the training set, and the remaining 17 cases are used as a test set. The training set and test set are ensured to cover nine syndromes, such as Heart Qi deficiency, Heart Yang deficiency, Spleen Yang deficiency, and Heart and Spleen deficiency, etc. And the diagnosis and recognition of syndrome are got through Naive Bayesian classification methods. The result shows that Naive Bayesian classification model improved by genetic algorithm not only has less desired characteristics, but also is more accurate in classification.

Wang Ruixiang [9] applied Bayesian network in TCM expert system. Through Bayesian network, syndrome differentiation of TCM, as the core of expert systems, determines whether there is a causal relationship between symptoms according to information entropy, and also determines the category of syndrome group. The advantage is that it treats the symptoms as an interconnected entirety.

Zhu Yonghua [10] constituted Bayesian network of syndrome differentiation in TCM, the node set including 916 symptoms in TCM system, 51 key syndrome elements and 1700 names of syndrome. The strength of association between each variable is set by “weight”, such as palpitations-Heart 50, Yang deficiency 20, Yin deficiency 20, Qi deficiency 15, and Blood deficiency 15; thus we can determine the causal relationship of symptoms and key syndrome elements. Using the established system, we can do data econometric analysis, verify the causal relationship of symptoms and key syndrome element, the relationship of key syndrome elements and names of syndrome by inference. The result is quite consistent with the TCM expert experience. All these shows that Bayesian network can be applied to the research of syndrome differentiation in TCM, thereby further quantify the syndrome differentiation system.

Zhang Lianwen [11] proposed a research method to analyse the objective and quantitative standard in TCM-Hidden Structure method. They believe that “waist and knee pain”, “red tongue little coating”, “thready pulse” and other symptoms which can be directly observed by the four diagnostic methods are significant variables; while so far, the syndrome of “Kidney Yin deficiency” which still can not be observed directly is the hidden variable. In addition to this syndrome, “Kidney Yin deficiency and dystrophy” and other features of the pathogenesis are also hidden variables. Relationship between hidden variables and hidden variables, and relationship between hidden variables and significant variables form a hidden structure. Using the hidden structure, the author well validated the Kidney deficiency model of TCM.

Wu Rong [12] formed a database by real-time collection of the information of famous TCM doctors’ diagnosis and treatment of Coronary heart disease and angina pectoris. They extract the key syndrome elements and related syndromes of Coronary heart disease and angina pectoris cases treated by famous TCM doctors through Bayesian network; express the contribution of each related syndrome in these doctors’ syndrome differentiation results in the form of conditional probability; reflect the essence of syndrome diagnosis; transform these doctors’ experiences into quantitative knowledge; and open up a new way to inherit these doctors’ experiences.

With the sample of TCM syndromes of depression patients, Qu Miao [13] developed the “Depression Syndrome Observation Table of TCM”; assessed the TCM symptoms, conducted the research on Bayesian network; and worked out the TCM syndromes type. The result shows that the syndrome type covers most symptoms of depression which include the core symptoms of depression and the different combinations of symptoms around, and reflects the changeable clinical syndrome of depression. So the research

on syndrome of TCM through Bayesian network is objective, scientific, and correspondent with TCM theory.

V. DISCUSSION

Bayesian network model can describe syndrome differentiation of TCM process which is of strong subjectivity and fuzziness in the form of intuitive graph and simple data. It provides a good methodology platform for the research on the standardization of TCM. The research of syndrome is one of the hotspots of TCM in recent years. The key syndrome element is the smallest unit of syndrome; syndrome differentiation is the process of combination of key syndrome elements; which fully reflects the flexibility of TCM. Bayesian network, as a hot research spot and an important achievement in artificial intelligence field, can be used for the analysis of complex multi-causality. The process of the reasoning based on probability calculation and mathematical derivation. It can be applied to fusion analysis of four diagnostic methods of TCM and the research of the complex causal relationship of symptoms and the relationship between the symptoms and key syndrome elements. Bayesian network model can simulate the learning and thinking process of human brain which is a good method to deal with data mining. It applies no limitation to the variables. As long as the syndrome differentiation data collected clinically is put into the computer, the model can start learning and reasoning by itself. And relative to human brain, it is objective. With the mass information of four diagnostic methods, Bayesian network integrates the prior knowledge and post-test data, and effectively handles the uncertain information in syndrome differentiation. And it takes a global point of view in the entire process. This is also consistent with the holistic thinking principle of TCM. Recent studies show that its reasoning results are quite consistent with the experience of TCM experts.

However, the establishment of Bayesian network comes from the data of clinical database. The reasoning level of the model lies on whether the information of cases is comprehensive and objective and whether the expert’s syndrome differentiation result is accurate. So it makes the demand on the quality of clinical data collected much more higher. Bayesian network is the algorithm based on frequency. In order to reduce the computation amount, some symptoms, key syndrome elements and names of syndrome with low frequency can not be included, which will definitely affect the accuracy of reasoning. The description of syndrome made by Bayesian network can only be qualitative not quantitative, so it is difficult to fully reflect the syndrome differentiation significance. Clinically, some symptoms are negatively correlated with syndrome differentiation result. But the local probability distribution parameter calculated by Bayesian network is without the difference of positive and negative. Therefore, the Bayesian network is a good method for information fusion analysis of four diagnostic methods, but it still can not fully reflect the thinking ability of syndrome differentiation in TCM.

Modernization and internationalization of TCM require the intercross of many subjects, the support cutting-edge technology, and constant practice and exploration. Research

of Bayesian network on data information fusion analysis of four diagnostic method of TCM has shown its great vitality, which is of great significance. As a new data mining technique, recent studies show the syndrome differentiation reasoning results of Bayesian network are consistent with the theory and clinics of TCM; and they are with certain objectivity. So there will be good development prospects for Bayesian network. It might be an effective method to breakthrough the standardization of syndrome differentiation in TCM by mining the mass information of four diagnostic methods in clinical database and establishing a TCM syndrome model through Bayesian network. Of course, the problems of how to provide higher sample quality, and improve the performance of syndrome classification and algorithm of Bayesian network still need constant exploration. With the in-depth research, we believe that Bayesian network will bring greater contributions to information fusion analysis of four diagnostic methods, so that further develop the standardization of syndrome differentiation in TCM.

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REFERENCES

- [1] JEN SEN F V. An introduction to Bayesian networks [M]. New York: Springer, 1996.
- [2] PEARL J. Probabilistic reasoning in intelligent systems [M]. San Mateo, CA: Morgan Kaufmann, 1988.
- [3] HECKERMAN, MAMDANIA, W ELLMAN M P. Real world application of Bayesian network [J]. Communication of ACM, 1995, 38: 24-26.
- [4] Wang Xuewei, Qu Haibin, Liu Xuesong, et al. Bayesian network approach to knowledge discovery in Traditional Chinese Medicine. Journal of Zhejiang University (Engineering Science), 2005, 39(7): 948-952.
- [5] Hai Xia. Discussion on new syndrome differentiating system by professor Zhu Wenfeng [N]. China News of Traditional Chinese Medicine (In Chinese), 2004, 5, Apr: 5.
- [6] Zhu Wenfeng, Yan Junfeng. The content and scientific significance of a new syndrome differentiating system based on new pattern element [J]. Medicine and Philosophy (In Chinese), 2005, 26(1): 69-71.
- [7] Liu Zhanqing. The research and implementation in the diagnosis of syndromes of TCM based on Bayesian classification algorithm. Science & Technology Information, 2007, 22: 148.
- [8] Zhang Liwei, Duan Chanlun, Xiong Zhiwei, et al. Study on the Application of Naive Bayesian Methods in Identifying Syndrome in TCM. Journal of Inner Mongolia University, 2007, 9, 38(5): 568-571.
- [9] Wang Ruixiang, Cui Lirui. The construction method of TCM experts system based on Bayesian network. China Medical Herald. 2007, 4(7): 58.
- [10] Zhu Yonghua, Zhu Wenfeng. Syndrome Differentiation System of Traditional Chinese Medicine Based on Bayesian Network. Journal of Hunan University (Natural Sciences), 2006, 33(4): 123-125.
- [11] Zhang Lianwen, Yuan Shihong. Hidden structure model and TCM syndrome differentiation-The basic structure and analytical tool of Hidden structure. Journal of Beijing University of TCM, 2006, 29(6): 365.
- [12] Wu Rong, Nie Xiaoyan, Wang Jie, et al. Study on the treatment of coronary heart disease differentiation rules by senior doctors of TCM based on Bayesian network. Chinese Journal of Information on TCM, 2010, 5(17): 598-599.
- [13] Qu Miao, Tang Qisheng, Bao Zuxiao, et al. The application research of Bayesian network model in TCM syndromes. Chinese Archives of Traditional Chinese Medicine, 2008, 7, 26(7): 1497-1498.